Elicitation-Induced Ambiguity Attitudes: Matching, Choice, and Cognitive Frictions*

Yuchi Li† Jilong Wu‡

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Abstract

We examine whether measured ambiguity attitudes depend on the elicitation method. Using a novel 10×10 symmetric stimulus design that presents identical binary choice problems either as standalone binary choices or embedded in matching probability (MP) and matching event (ME) lists, we document three findings. First, MP produces substantially stronger a-insensitivity than binary Choice, with half of the individual-level divergence constituting outright preference reversals; data quality metrics systematically favor the Choice method. Second, ME produces a-oversensitivity — the precise sign reversal of the MP pattern — despite presenting the same choice problems, directly refuting the invariance prediction of standard preference-based theories. Third, a two-stage noisy coding model adapted to the ambiguity domain organizes all findings under a single mechanism: the direction and magnitude of elicitation-induced distortion depend on whether the fixed first-stage option is ambiguous or risky. These results call into question the reliability of matching-based ambiguity attitude estimates and argue for binary choice as the preferred elicitation instrument.

Keywords: Ambiguity attitudes; a-insensitivity; matching probability; binary choice; noisy coding; procedural invariance

JEL Classification: C81; C91; D81; D91

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†Yuchi Li: Risl$\alpha\beta$, Department of Economics, Gent University, yuchili@ugent.be.

‡Jilong Wu: Risl$\alpha\beta$, Department of Economics, Gent University, jilongwu@ugent.be.

1 Introduction

Many economically consequential decisions involve not merely quantifiable probabilities but genuine uncertainty about the probability distribution itself. Ellsberg (1961) demonstrated how DMs systematically deviate from expected utility predictions: in one of his thought experiments, most people prefer drawing from an urn with a known equal proportion of red and black balls over an urn whose composition is entirely unknown — even though both urns could in principle be equally favorable. This preference for known over unknown probabilities, now widely known as *ambiguity aversion*, is fundamentally at odds with the subjective expected utility framework (Savage, 1954) and spurred a class of decision theories that model ambiguity attitudes through a second-order transformation over utilities or expected utilities, usually referred to as *outcome-driven* models.[1] Subsequently, the finding that ambiguity attitudes are probability-dependent — with *ambiguity seeking* prevailing for unlikely events (Trautmann and van de Kuilen, 2015) — motivated the notion of ambiguity-generated likelihood insensitivity (*a-insensitivity*, Dimmock et al., 2016; Baillon et al., 2018) and increased the appeal of other models that capture ambiguity attitudes through a non-additive weighting of events or sources, usually referred to as *event-driven* models.[2]

In this paper, we experimentally show that the pronounced ambiguity seeking and strong a-insensitivity documented in recent studies depend importantly on the use of the matching probability (MP) elicitation method (Kahn and Sarin, 1988; Dimmock et al., 2016; Baillon et al., 2018). When instead DMs face the same decision problems as standalone choices, ambiguity seeking is substantially attenuated and a-insensitivity is markedly weaker. This divergence constitutes strong evidence that these patterns are shaped at least in part by elicitation-

[1] Important examples include maxmin expected utility (Gilboa and Schmeidler, 1989), α-maxmin expected utility (Ghirardato et al., 2004), and the smooth ambiguity model (Klibanoff et al., 2005).

[2] Important examples include Choquet expected utility (Schmeidler, 1989), the source-dependent generalization of prospect theory (Tversky and Kahneman, 1992; Tversky and Wakker, 1995), neo-additive capacities (Chateauneuf et al., 2007), and source theory (Abdellaoui et al., 2011; Baillon et al., 2025).

dependent cognitive and response processes, since the two methods draw on an identical underlying problem set. We offer a cognitive account of why the two formats generate systematically different ambiguity attitudes — an account that in turn yields a sharp additional prediction: transforming MP into its natural counterpart — the matching event (ME) method — should produce a pattern precisely opposite to that under MP, even when the underlying problem set is held constant. We design and implement a novel experimental test of this prediction and find strong support for it.

Understanding ambiguity attitudes is central across many economic domains. Empirically, ambiguity attitudes, and a-insensitivity in particular have been linked to consequential real-world behaviors, including stock market non-participation, limited insurance uptake, and reduced entrepreneurial activity (see the early review in Trautmann and van de Kuilen, 2015 and recent evidence in Bianchi and Tallon, 2019; Belissa et al., 2020; Anantanasuwong et al., 2024). Reliable measurement of these attitudes is therefore a prerequisite for drawing valid inferences about their economic implications. Our results, however, suggest that commonly used matching-based elicitation methods introduce systematic variation in measured behavior that is not straightforwardly attributable to stable ambiguity attitudes alone. If the magnitude of measured a-insensitivity or ambiguity attitudes depends on the elicitation format, then estimates obtained from different methods need not be directly comparable, and correlations with field behavior may partly reflect features of the elicitation procedure rather than the underlying construct alone. This renders cross-study comparisons of effect sizes difficult to interpret and potentially obscures the true economic relevance of ambiguity attitudes.

Beyond measurement, the exact shape of ambiguity attitudes is theoretically consequential. Descriptively, the coexistence of ambiguity aversion for likely events and ambiguity seeking for unlikely events appears to be the dominant empirical pattern (see Trautmann and van de Kuilen, 2015). This pattern poses a challenge for several leading models of ambiguity. For example, as noted by Machina (2014), the smooth ambiguity model (Klibanoff et al., 2005) cannot simultaneously gen-

erate ambiguity aversion for likely events and ambiguity seeking for unlikely ones (see also Baillon and Bleichrodt, 2015). More importantly, our findings of procedural invariance violation raise a distinct question: whether elicited ambiguity attitudes can be treated as stable, method-invariant primitives at all. Our results suggest instead that the apparent preference structure of ambiguity attitudes may itself be generated by elicitation-induced cognitive processes, rather than merely revealed by measurement.

Matching versus Choice. Our first contribution is a new experiment that carefully tests for and compares the qualitative incidence and quantitative intensity of ambiguity attitudes and a-insensitivity across two canonical elicitation environments. First, we elicit participants' *matching probabilities* for a series of ambiguous events via choice lists, following the standard practice in the literature (e.g., Baillon and Bleichrodt, 2015; Baillon et al., 2018). We refer to this as the *matching* task. The matching probability of an event is defined as the probability at which an individual is indifferent between a lottery that pays the prize with that probability and a lottery that pays the prize conditional on the event. Second, we decompose the individual binary choices contained in the MP lists into standalone choices between the risky and the ambiguous lottery, presented one by one on separate pages. We refer to this as the *Choice* task. Because both tasks present an identical set of underlying problems, this design maximizes comparability across elicitation formats and provides the strongest possible test of the hypothesis that the two methods yield consistent attitudes. Indeed, under standard ambiguity models, the two tasks are decision-theoretically equivalent. Any systematic difference in elicited attitudes therefore points to elicitation-dependent cognitive or response processes, rather than to stable underlying preferences alone.

Nonetheless, we document statistically and economically significant differences in elicited ambiguity attitudes across the two tasks. In the Matching task, we replicate the conventional pattern of a-insensitivity widely reported in the ambiguity literature: ambiguity seeking for unlikely events and ambiguity aversion for likely ones, with elicited matching probabilities closely aligned with those in highly com-

parable studies (e.g., Dimmock et al., 2015, 2016). In the Choice task, we observe a qualitatively similar pattern, but the magnitude of ambiguity non-neutrality is reduced by approximately 50%. More strikingly, a substantial share of participants (approximately two thirds) exhibit unstable responses across tasks. Among participants exhibiting instability, roughly two thirds are ambiguity seeking for unlikely events ($\hat{p} = 0.1$ and $\hat{p} = 0.2$) in the Matching task yet ambiguity averse for the same events in the Choice task. This reversal pattern indicates that the divergence between elicitation formats is not merely a rescaling of the same underlying attitude profile: for a substantial fraction of participants, the sign of the measured attitude changes entirely across methods, reflecting a qualitative inconsistency rather than a uniform compression.

Beyond the attenuation and reversal findings, we document several further results concerning the data quality within each task. First, among the 100 underlying problems, ten involve a dominant ambiguous option — namely, those in which the ambiguous lottery is degenerate, guaranteeing a prize regardless. With the exception of the row in which the risky option is equally degenerate, rational DMs should unambiguously prefer the ambiguous option irrespective of their ambiguity attitudes. Yet participants commit significantly more violations of this dominance criterion in the Matching task than in the Choice task. Second, for the repeated decision problems included as consistency checks, participants are substantially more likely to give identical answers across repetitions in the Choice task than in the Matching task. Third, participants exhibit lower between-list consistency in the Matching task than in the Choice task — the requirement that matching probabilities be non-decreasing across ambiguous events of increasing likelihood is violated more frequently under the list format. Taken together, these results point in the same direction: relative to the Choice task, the Matching task elicits noisier and less consistent responses, with more frequent violations of basic rationality benchmarks.

Theoretical Explanation and Experimental Validation. We propose and experimentally test an explanation grounded in the idea that the observed pat-

terns reflect cognitive frictions inherent to the information-processing demands of each elicitation procedure. Our account draws on a growing class of noisy coding models, which explain distortions in likelihood-dependent behavior as arising not from stable preferences but from the imprecise, Bayesian way in which DMs internally represent probabilistic information (Frydman and Jin, 2022; Vieider, 2024; Bouchouicha et al., 2025; Khaw et al., 2025; Vieider, 2025). Under this view, DMs combine a noisy perception of the true winning odds with a prior expectation in a Bayesian fashion, compressing internal representations toward the prior and thereby generating probability distortions endogenously. We extend this framework to the domain of ambiguity by building on Vieider (2025): because an ambiguous event is inherently harder to represent mentally than a risky one, it is encoded with greater noise and receives strictly less weight relative to the prior. This asymmetry in encoding precision generates a-insensitivity endogenously, without any additional preference parameters.

The procedural architecture of each elicitation method then determines how severely this compression manifests. In binary choice, DMs simultaneously compare the perceived log-odds of the risky and ambiguous prospects; because the two prospects are evaluated separately, the distortions in their internal representations partially offset each other, yielding null to mild a-insensitivity — consistent with our Choice results. The Matching Probability method, by contrast, imposes a sequential cognitive procedure: the DM must first form an evaluation of the fixed ambiguous prospect and then search for the objective probability that equates to it (Bouchouicha et al., 2025; Khaw et al., 2025). This two-stage structure compounds the compression, since the Stage 1 anchor is already shrunk toward the prior before Stage 2 calibrates the matching probability against it — producing stronger a-insensitivity and lower response quality than binary choice.

The same logic generates a sharp prediction for the Matching event method, where the roles of the two prospects are exchanged: fixing the risky prospect in Stage 1 and searching for a matching ambiguous event reverses the direction of the distortion, yielding a-oversensitivity. Moreover, because the fixed risky option

is encoded with greater precision than the fixed ambiguous option, the first-stage anchor is less severely compressed under ME than under MP — implying that the magnitude of a-oversensitivity under ME should fall short of the magnitude of a-insensitivity under MP. We experimentally test and find strong support for both predictions.

Related Literature. Among these noisy coding works, the closest paper to ours is Bouchouicha et al. (2025), who compare certainty equivalents elicited via multiple price lists — where DMs choose between a fixed two-outcome lottery and a sequence of varying sure amounts — against binary choices extracted from the constituent rows of those same lists. They document the fourfold pattern of risk attitudes under the list format but find that it largely collapses under binary choice, with DMs exhibiting more uniform risk aversion for gains. They account for this divergence through a noisy coding framework in which the sequential evaluation structure of the list format amplifies cognitive distortions relative to the simultaneous comparison in binary choice. Our study is directly inspired by their design and theoretical framework, but differs from it in several respects that constitute independent contributions.

First, and most substantively, we extend the cognitive-frictions account to the domain of ambiguity, where the informational asymmetry between risky and ambiguous prospects introduces a new theoretical dimension absent from the pure-risk setting. Second, our design elicits both matching tasks and binary choices from the same participants, enabling within-subject comparisons and direct evidence on preference reversals across methods — an analysis that between-subject designs preclude. Third, the symmetric structure of our stimuli allows us to introduce the ME elicitation as a natural counterpart to the standard MP elicitation. Because ME exchanges the fixed and varying roles of the two lotteries, it generates a qualitatively distinct and testable prediction — a-oversensitivity rather than a-insensitivity — that would not arise in a risk-only setting and that provides a particularly clean experimental test of the noisy coding mechanism as applied to ambiguity.

More broadly, our paper contributes to a large body of work measuring ambiguity attitudes via matching probabilities (Kahn and Sarin, 1988; Dimmock et al., 2016; Baillon et al., 2018; Kocher et al., 2018; von Gaudecker et al., 2025). Across this literature, a clear pattern emerges: DMs exhibit a-insensitivity, with matching probabilities compressed toward fifty-fifty relative to the ambiguity-neutral probabilities, and the combination of ambiguity seeking for unlikely events and ambiguity aversion for likely events is routinely documented as the modal finding.[3] Evidence from designs that elicit ambiguity attitudes on unlikely events through direct binary choice is comparatively sparse, and what exists suggests a meaningfully different picture. The first-round choice in a bisection procedure offers a partial approximation, yet the magnitude of ambiguity seeking for unlikely events is already noticeably weaker than under full matching protocols: in Dimmock et al. (2016), for instance, fewer than half of participants select the ambiguous option at $p = 0.1$. Tracing back further, Einhorn and Hogarth (1986) provide evidence that under direct pairwise choice the proportion of ambiguity-seeking responses shrinks substantially, and the modal response for very unlikely events is ambiguity aversion — a finding broadly consistent with our binary Choice results.

Our paper also relates to a long literature testing *procedural invariance* in decision-making under risk (e.g., Dave et al., 2010; Loomes and Pogrebna, 2014; Pedroni et al., 2017; Holzmeister and Stefan, 2021; Friedman et al., 2022), where a recurring finding is that measured risk attitudes are sensitive to the elicitation method employed. As nearly all existing studies focus on risky choice, extending this investigation to the domain of ambiguity with a clean and systematic comparison between the Choice method and the Matching method constitutes a further contribution of this paper. On this front, Maafi (2011) and Trautmann et al. (2011) are the only two studies we are aware of that directly examine preference reversals under ambiguity.

[3]Some studies elicit matching probabilities via a bisection procedure, which presents binary choices sequentially across pages. Despite the superficially binary format of each individual problem, Bouchouicha et al. (2025) argue that the overall procedure more closely resembles a matching task than a genuine binary choice, inasmuch as each response is conditioned on the preceding one — thereby mimicking the sequential evaluation structure of a list.

Trautmann et al. (2011) is more comparable to our study. They find that approximately 30% of participants choose the ambiguous lottery over the risky counterpart in a binary choice, while about half of that group instead assign a higher willingness to pay to the risky lottery in a valuation task — a pattern indicating inconsistency across methods. Instructive as this work is, its design leaves two limitations for our purposes. First, their study focuses on a single moderate likelihood, a restriction that precludes a richer characterization across the full likelihood range. Second, a single binary choice reveals only the ordinal direction of ambiguity preference, whereas a matching method yields a cardinal measure of the indifference point; the two are therefore not directly comparable as measures of magnitude. Our design addresses these limitations directly: by using an identical problem set across both elicitation methods and spanning the full range of likelihoods, we document a richer and more systematic picture of ambiguity preference reversals. In particular, we find that the direction and magnitude of such reversals depend critically on the likelihood involved (Feldman and Ferraro, 2025) — a dependence that Trautmann et al. (2011) could not detect. Our findings are more consistent with an alternative interpretation than with their reference-dependence account, which does not readily explain why the direction of reversals varies systematically with the likelihood level.

2 Choice versus Matching Probability

In this section, we will focus on the experimental investigation concerning whether decision makers (DMs) behave systematically differently across two common ambiguity attitude elicitation methods — binary *Choice* and *Matching* probability. Except for the aggregate level analysis of the overall pattern, our individual analysis will provide fresh and complete evidence on preference reversals under ambiguity (Maafi, 2011; Trautmann et al., 2011). Last, we will further discuss choice sub-optimality and inconsistency, which offers a powerful perspective for distinguishing our noisy coding model from existing ambiguity models (e.g., Klibanoff

et al., 2005; Baillon et al., 2025).

2.1 Implementation

Sample and Incentives. The experiment was conducted on Prolific, using a demographically representative sample of the United States adult population. A total of 400 valid responses were collected.[4] The median completion time was approximately 30 minutes, with a median age of 47 and 198 female participants. Participants received a fixed participation fee of $6. Monetary incentives for the decision tasks were implemented via a between-subject random incentive system: participants were informed that one in every ten participants would be randomly selected to have one of their choices played out for real.[5]

Procedure. Participants were randomly assigned to either the matching probability condition (MP, $n = 201$) or the matching event condition (ME, $n = 199$). We apply an identical experimental design and procedure to both conditions, and the only difference is the format of matching. In this section, we will concentrate on the MP condition.

After reading the experiment instructions and completing a set of training questions, participants were required to pass three comprehension check questions before proceeding to the choice tasks. To encourage effort, an additional $0.15 was awarded for each question answered correctly on the first attempt.[6] Each participant completed two choice tasks: the *Matching* task and the binary *Choice* task. The order of the two tasks was randomized and balanced across conditions. Between the two tasks, participants completed 15 cognitive ability test questions,

[4]We excluded 9 whose experiment duration fell below 800 seconds, a threshold used to screen out inattentive responses.

[5]This between-subject variant of the random incentive system is a well-established practice in large-scale online experiments, preserving incentive compatibility while keeping implementation costs tractable (e.g., Abdellaoui et al., 2011; Baillon et al., 2018). Importantly, participants were informed at the very beginning of the experiment that both their bonus eligibility and the ambiguous source (i.e., the opaque box's composition) had been predetermined by the computer. Beyond enhancing trust in the procedure, this approach theoretically prevents strategic hedging across choices (Baillon et al., 2022).

[6]86% of participants passed all three comprehension questions correctly on their first attempt.

Figure 1: An example of a binary choice problem. Notes: The left (risky) box yields an objective winning probability of $p = 0.5$. The right (ambiguous) box yields an unspecified winning probability that could be anywhere in $[0, 1]$. Both options pay $20 for a winning draw and nothing otherwise.

serving as a midway break. Upon completing both choice tasks, participants filled out a brief demographic questionnaire to conclude the experiment.

2.2 Experimental stimuli and methods

Basic Problem. Our experiment employs a classic two-urn design inspired by Ellsberg (1961). Each decision problem presents participants with two *boxes*, each containing exactly ten balls numbered 0 through 9. Participants are asked to choose one box from which a ball will be drawn at random, with a constant monetary payoff of $20 determined by whether the drawn ball falls within a pre-specified winning range.[7] The two boxes differ in one critical dimension: the composition of balls in the *risky* box is fully transparent to the participant, whereas the composition of balls in the *ambiguous* box is concealed. Figure 1 provides an example of such decision problems.

Stimuli. The full stimulus set is defined by the Cartesian product of the two boxes' winning ranges. Each box admits ten possible winning ranges — 0, 0–1, 0–2, . . . , 0–9 — yielding a total of $10 \times 10 = 100$ distinct binary choice problems.

Binary Choice. In the binary Choice task, participants were asked to answer binary choice between a risky lottery and an ambiguity lottery. Each decision

[7] Throughout the whole experiment, we avoid using any term like "lottery" or "bet" to ensure a neutral communication with participants.

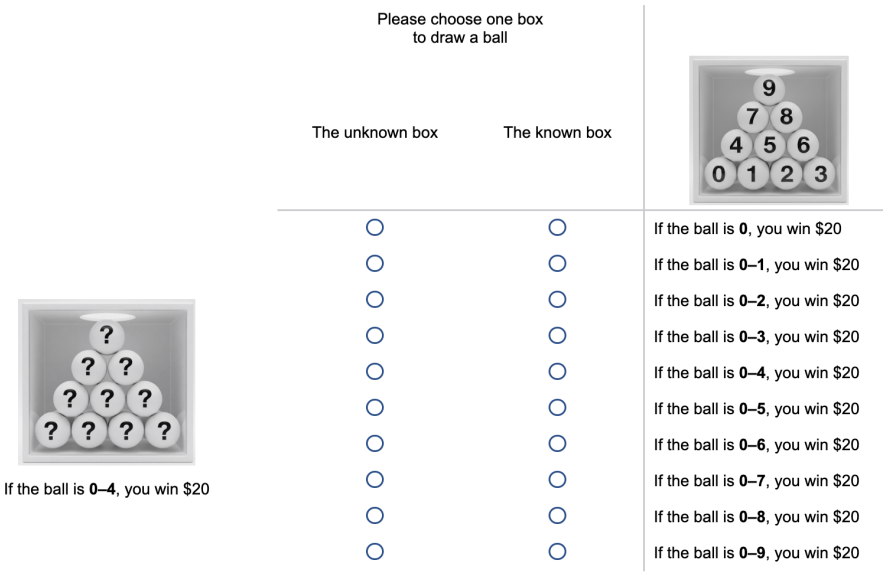

Figure 2: Screenshot of the ascending matching probability choice list interface. Notes: Each row presents a binary choice between the ambiguous box (left, fixed across rows) and a risky box (right) whose winning range expands monotonically from a single ball (0) at the top to all ten balls (0–9) at the bottom. The corresponding objective winning probabilities range from $p = 0.1$ to $p = 1.0$ in increments of 0.1.

problem was displayed on a separate page, presented in a fully randomized order across all 100 problems. Figure 1 provides a screen shot of this Choice task.

Matching Probability. In the Matching task, the matching probability mp is elicited via *choice lists*, the most common method among behavioral economists to derive a matching value (Abdellaoui et al., 2011; Baillon et al., 2018). Specifically, a MP choice list consists of a fixed ambiguous lottery on its left side and a vector of 10 risky lotteries on its right side. The winning range of the risky lottery monotonically changes from the top to the bottom in a step of one. In total, 10 MP choice lists were displayed in a random order. Besides, we also randomize the (ascending or descending) sequence inside lists in a between-subject way.[8] Figure 2 provides an example of (ascending) MP lists.

Comparability. The key to our design is that we make participants answer an identical stimuli set across both tasks. Thus, from a payoff perspective, subjects make identical decisions under an identical payoff scheme in our Matching and

[8]This practice mitigates specific context effects while keeping participants' cognitive burden manageable.

Choice tasks. This constitutes a particularly stringent test of any preference-based theory (e.g., Klibanoff et al., 2005; Baillon et al., 2025) which predicts identical behavioral patterns across two tasks. Additionally, the rich stimuli set also allows us to have the full picture of ambiguity attitudes and explore the procedural invariance throughout the whole range from small likelihood to large likelihood.

2.3 Results

Following conventional practice (Kahn and Sarin, 1988; Dimmock et al., 2016), we first derive *event-specific* ambiguity attitude (AA) indexes. Each index $AA_{\hat{p}}$ captures the level of "local" ambiguity aversion for an ambiguous event whose objective winning probability $\hat{p}$ would be used by an ambiguity-neutral decision maker — hereafter referred to as the *ambiguity-neutral (or a-neutral for short) likelihood*. Formally,

$$AA_{\hat{p}} = \hat{p} - mp(\hat{p}),$$

where $mp(\hat{p})$ denotes the matching probability elicited for the ambiguous event with a-neutral likelihood $\hat{p}$.[9] A positive (negative) value of $AA_{\hat{p}}$ indicates ambiguity aversion (seeking) at likelihood $\hat{p}$, with the elicited matching probability falling below (above) the a-neutral benchmark. The a-insensitivity pattern corresponds to a configuration in which $AA_{\hat{p}}$ is negative for small likelihoods — reflecting ambiguity seeking — and positive for moderate to large likelihoods — reflecting ambiguity aversion — tracing out an inverse-S shaped profile across the probability range.

To complement the event-specific analysis, we also compute two global indexes following Dimmock et al. (2016). For each participant, a best-fitting line $mp(\hat{p}) \mapsto c + s\hat{p}$ is estimated between the matching probabilities $mp(\hat{p})$ and the a-neutral likelihoods $\hat{p}$ across the full probability range. The global index of a-insensitivity is defined as $a = 1 - s$, capturing the flatness of the matching prob-

[9] We use the midpoint of the switching range, approximated by the choice proportion, to derive matching probabilities. Specifically, for an ambiguous lottery $(\$20, E; 0)$, if a participant selects the ambiguous option in 3 out of the 10 corresponding rows, her matching probability for event E is recorded as 0.35.

ability profile — a tendency to compress all likelihoods toward 50-50. The global index of ambiguity aversion is defined as $b = 1 - s - 2c$, capturing the average elevation of the profile relative to the ambiguity-neutral benchmark. By construction, a and b are orthogonal, summarizing the curvature and level of the matching function respectively, and a positive value of each index indicates a-insensitivity and ambiguity aversion.

Aggregate Analysis. Figure 3 plots the event-specific ambiguity attitude indexes $AA_{\hat{p}}$ across all nine a-neutral likelihoods for both elicitation formats. Under the Matching format, $AA_{\hat{p}}$ is substantially negative for low likelihoods — reaching approximately -0.14 at $\hat{p} = 0.1$ — crosses zero near $\hat{p} = 0.4$, and rises to roughly 0.22 at $\hat{p} = 0.9$, closely replicating the event-specific indexes reported by Dimmock et al. (2016). This systematic sign reversal — ambiguity seeking for unlikely events and ambiguity aversion for likely ones — is the hallmark of a-insensitivity: decision makers fail to discriminate sufficiently among likelihood levels, compressing their subjective likelihood judgments toward fifty-fifty (Dimmock et al., 2016; Baillon et al., 2018). Under the Choice format, the $AA_{\hat{p}}$ indexes lie systematically closer to zero across virtually the entire likelihood range. Both tails are attenuated by roughly 50% — weaker ambiguity seeking at low likelihoods ($AA_{0.1}^{\text{Choice}} = -0.06$ *vs.* $AA_{0.1}^{\text{Matching}} = -0.14$) and weaker ambiguity aversion at high likelihoods ($AA_{0.9}^{\text{Choice}} = 0.12$ *vs.* $AA_{0.9}^{\text{Matching}} = 0.22$) — yielding a flatter overall profile. This symmetric compression is precisely the signature of reduced a-insensitivity rather than a uniform shift in the level of ambiguity aversion per se.

Individual Analysis. The within-subject design allows us to compare $AA_{\hat{p}}$ across the two elicitation methods at the individual level. For events with small a-neutral probability — $\hat{p} = 0.1$ and $\hat{p} = 0.2$ — only 68 and 57 out of 201 participants, respectively, report identical $AA_{\hat{p}}$ under both methods, meaning that over two thirds exhibit inconsistent attitudes. As reported in the panel A of Table 1, among those who disagree, 62.4% (56.9%) assign a higher $AA_{\hat{p}}$ under the Matching method

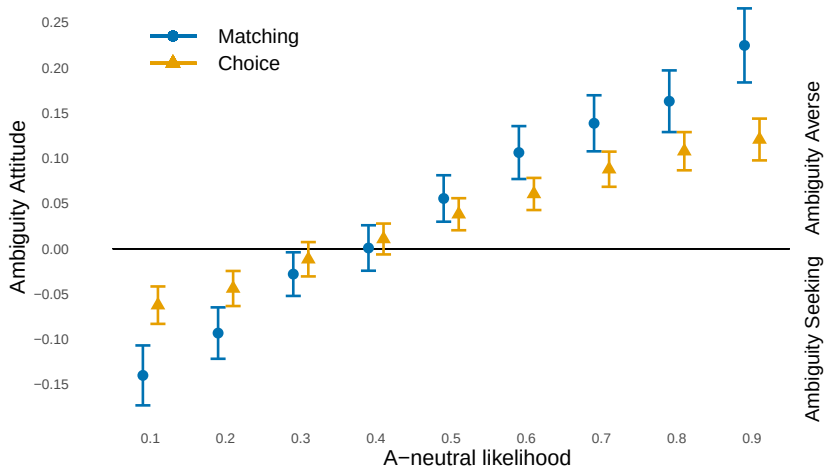

Figure 3: Event-Specific Ambiguity Attitude Indexes by Response Format. Notes: Matching denotes the choice list format and Choice denotes the binary choice format. Positive (negative) values indicate ambiguity aversion (seeking). Error bars denote 95% confidence intervals. See corresponding statistics in Table B.1.

Table 1: Instability and preference reversal by event likelihood.

	Panel A: Instability			Panel B: Reversal		
$\hat{p}$	$AA^{\text{Matching}} < AA^{\text{Choice}}$	$AA^{\text{Matching}} = AA^{\text{Choice}}$	$AA^{\text{Matching}} > AA^{\text{Choice}}$	No Reversal	PR1	PR2
0.1	50	68	83	129	41	31
0.2	62	57	82	130	48	23
0.8	85	44	72	142	42	17
0.9	93	41	67	151	34	16

Notes: $\hat{p}$ denotes the event's a-neutral probability. $N = 201$ per $\hat{p}$ level. Panel A reports the number of participants whose AA derived from Matching is lower than, equal to, or higher than that derived from Choice. Panel B classifies participants by preference reversal type: No reversal denotes participants whose AA carries the same sign across both methods; PR1 ($AA^{\text{Matching}} < 0 < AA^{\text{Choice}}$) denotes participants who are ambiguity seeking under Matching but ambiguity averse under Choice; PR2 ($AA^{\text{Choice}} < 0 < AA^{\text{Matching}}$) denotes the opposite pattern. Full results across all A-neutral probability levels are reported in Tables B.2 and B.3.

than under the Choice method. The pattern reverses for large a-neutral probabilities: at $\hat{p} = 0.8$ and $\hat{p} = 0.9$, the number of consistent participants drops further — to only 44 and 41 out of 201 — with approximately four fifths exhibiting inconsistent attitudes, and 54.1% (58.1%) of them now assigning a higher $AA_{\hat{p}}$ under the Choice method than under the Matching method.

To examine this more closely, we further classify participants by their ambiguity attitude reversal type (see the panel B of Table 1). Overall, approximately 31.9% of participant-event observations exhibit a preference reversal across the

Table 2: Summary statistics of a-insensitivity and ambiguity aversion indices by elicitation method.

	A-insensitivity index a			Ambiguity aversion index b			
Method	Mean	Median	SE	Mean	Median	SE	N
Matching	0.431	0.297	0.032	0.091	0.044	0.018	201
Choice	0.231	0.145	0.020	0.066	0.056	0.013	201

Notes: For a-insensitivity index $a = 1 - s$ and the ambiguity aversion index $b = 1 - s - 2c$, s and c are the slope and intercept of a best-fitting line $mp(\hat{p}) \mapsto c + s\hat{p}$ estimated between each participant's matching probabilities and the corresponding A-neutral likelihoods. A positive a indicates a tendency to compress likelihoods toward 50-50; a positive b indicates ambiguity aversion on average. SE denotes the standard error of the mean.

two methods, indicating that a non-trivial share of participants change the *sign* of their $AA_{\hat{p}}$ depending on how preferences are elicited. Preference reversals are most frequent at low a-neutral probability levels, reaching 35.8% at $\hat{p} = 0.1$ and declining monotonically to 24.9% at $\hat{p} = 0.9$, suggesting that the two methods diverge most sharply when the underlying event is unlikely. Among those who exhibits such a preference reversal, for small likelihoods, the dominant type consists of participants who are ambiguity seeking under the Matching method but ambiguity averse under the Choice method, accounting for 65.2% of all reversal cases — consistent with the aggregate findings above. Taken together, these results indicate that the method effect is not simply a uniform attenuation of ambiguity non-neutrality, but reflects a more fundamental instability with substantial individual-level heterogeneity. Spearman rank correlations of $AA_{\hat{p}}$ across methods corroborate this conclusion: the overall coefficient is only 0.347, falling as low as 0.157 ($p = 0.026$) for the high-likelihood event at $\hat{p} = 0.9$.

Table 2 reports the summary statistics of the two global indexes across elicitation methods. Both the a-insensitivity index a and the ambiguity aversion index b are positive under both Matching and Choice, indicating that participants exhibit likelihood insensitivity and ambiguity aversion on average regardless of how preferences are elicited. Notably, the estimates of a and b under Matching (0.43 and 0.09) closely replicate those reported in Dimmock et al. (2016) (0.41 and 0.12). Turning to the cross-method comparison, the ambiguity aversion index b does not

differ significantly across the two methods (0.091 *vs.* 0.066; $p > 0.1$), suggesting that the average elevation of the matching function is robust to the elicitation format. By contrast, the a-insensitivity index a is substantially larger under Matching than under Choice (0.43 *vs.* 0.23, $p < 0.01$), echoing the stronger a-insensitivity pattern we document in the event-specific analysis.[10] Finally, the cross-method Spearman correlations reveal a moderate agreement for the ambiguity aversion index b ($\rho = 0.49$) but a substantially lower agreement for the a-insensitivity index a ($\rho = 0.20$), consistent with the finding that preference reversals are concentrated at the likelihood extremes where the a-insensitivity signature is most pronounced.

Data Quality. The low cross-method correlation in responses — particularly in the a-insensitivity index aa a — raises the question of whether one method produces noisier data than the other. We address this by examining three complementary metrics. Given that the format of choice list visually organize those binary choices and group them by the ambiguous lotteries, some may argue that the structured list format helps participants organize their choices, thereby yielding cleaner preference measures. In contrast, others would say that binary choice can better capture preferences given its simplicity and free from strategic complexity. Our analysis reveals the difference in data quality and confirms that the binary Choice method produces less decision noise.

We first leverage those choices between a degenerate lottery and a nondegenerate lottery to examine the direct stochastic dominance violation. Specifically, when the ambiguous lottery is degenerate, out of the nine choices, participants choose the dominated risky one for 2.30 times when they answer in a choice list, compared to 1.04 times when they answer in separate pages ($p < 0.01$). Similarly, when the dominant lottery is risky, participants choose the dominated ambiguous one for 1.04 times in the Matching method, compared to the lower figure of 0.38 in the Choice method ($p < 0.01$).

[10]The results are robust to our random-effect regression after controlling several personal variables (see Table B.4). Interestingly, the regression results reveal that the a-insensitivity index a negatively correlates with individual cognitive ability, while the ambiguity aversion index b is higher among females.

Our experiments also repeat a MP choice list whose ambiguous lottery has an a-neutral likelihood equal to 0.2 in the Matching task and repeat the corresponding 10 binary choices in the Choice task. We use Fisher's exact test to compare the switching rate — the proportion of participants whose choice reverses between the two repetitions — across the ten known probability levels. The two methods exhibit comparable stability at low likelihood levels ($\hat{p} \leq 0.3$), but Choice produces significantly fewer switches than Matching at intermediate to high likelihood levels ($\hat{p} = 0.4$ to 0.9, all $p < .05$), with the gap widening as $\hat{p}$ increases.[11]

A complementary perspective on data quality comes from examining multiple switching behavior within the 10×10 choice matrix. Each column constitutes an MP choice list, in which the varying option's attractiveness changes monotonically as the participant moves down the list; a rational participant should therefore switch at most once per column. We refer to violations of this as *within-list* multiple switching. Each row does not form an explicit list, but the same logic applies horizontally: the fixed option is the risky lottery while the ambiguous lottery varies, and a consistent participant should likewise switch at most once per row. We refer to violations of this as *between-list* multiple switching. As expected, within-list multiple switching is more prevalent under the Choice method than under the Matching method (1.67 *vs.* 1.16, $p < 0.01$), since the explicit list format nudges participants toward a single switch. More revealing is the between-list pattern: multiple switching is significantly more frequent under the Matching method than under the Choice method (3.5 *vs.* 1.86, $p < 0.01$), suggesting that the list structure suppresses within-list inconsistency at the cost of distorting responses more broadly across lists.

Across all three data quality metrics — stochastic dominance violations, test-retest reliability, and between-list consistency — the evidence consistently points in the same direction: the Choice method elicits cleaner and more internally consistent responses than the Matching method. This is consistent with the interpretation that elevated cognitive noise in the Matching method, induced by its

[11] This result is robust to Cohen's κ, which corrects agreement rates for chance. See Table B.5 for full details.

list structure, is at least partly responsible for the stronger a-insensitivity pattern documented above — a mechanism we formalize in the theoretical section that follows.

3 A Noisy Coding Account

The results documented above present a fundamental challenge to standard preference-based theories of ambiguity. Under any standard preference-based model — including Choquet expected utility (Schmeidler, 1989), maxmin expected utility (Gilboa and Schmeidler, 1989), the smooth ambiguity model (Klibanoff et al., 2005), prospect theory (Tversky and Kahneman, 1992), and source theory (Abdellaoui et al., 2011; Baillon et al., 2025) — elicited ambiguity attitudes must be invariant to the presentation format of decision problems, provided the underlying choice problems are identical. This is because these models root ambiguity attitudes entirely in the primitives of the act being evaluated — the event, the prize, and the decision maker's preferences — none of which vary across the Matching and Choice methods in our experiment. The dramatic attenuation of a-insensitivity, the high rate of individual-level reversals, and the systematic differences in data quality we document therefore cannot be rationalized by any such theory. Along with the accumulating evidence of context- and procedure-dependence in both risk and ambiguity attitudes (e.g., Trautmann and van de Kuilen, 2015; McGranaghan et al., 2024; Bouchouicha et al., 2025), our results point to the need for theories that explicitly model the cognitive processes generating observed choices, rather than treating preferences as fixed primitives that elicitation merely reveals.

We propose such an account, building on the noisy coding framework recently developed for decision under risk by Bouchouicha et al. (2025) and extended to the domain of ambiguity by Vieider (2025). The technical arguments underlying the implications we derive require significant notation and are deferred to Appendix A. We stress here that these implications do not depend on the idiosyn-

crasies of our modeling approach as originally proposed by Bouchouicha et al. (2025), but rather appear to be general implications of noisy coding applied to choice-like and matching-like evaluation procedures. This conclusion is broadly consistent with related noisy coding frameworks developed in the risk domain: Khaw et al. (2025), in contemporaneous work, draw similar conclusions about valuation-like procedures using a noisy coding setup that builds on the somewhat different modeling choices in Khaw et al. (2021), suggesting that the mechanism generalizes across modeling approaches.

The core insight of noisy coding is that decision makers do not perceive probabilities or event likelihoods precisely; instead, they encode them with noise and compensate for this imprecision by combining their noisy perception with a prior expectation in a Bayesian fashion. In a basic setup involving a binary choice between a risky lottery and an ambiguous lottery, Vieider (2025) formalizes this for the ambiguity domain by modeling a DM's beliefs about the winning probability of an ambiguous urn as a Beta distribution $p_a(x) = \mathrm{Be}(\alpha_a, \beta_a)$, capturing both a central tendency and the precision of that estimate. The key asymmetry is that the variance of the ambiguous belief distribution exceeds that of the corresponding risky one $p_r(x) = \mathrm{Be}(\alpha_r, \beta_r)$, reflecting the DM's lower confidence in an unknown probability. The DM then infers the effective probability by shrinking her noisy perception toward a prior, yielding a compressed subjective likelihood scale. This Bayesian shrinkage produces a-insensitivity endogenously — and at a modest level of noise, this is precisely what the Choice method reveals: $AA_{0.1}^{\text{Choice}} = -0.06$ and $AA_{0.9}^{\text{Choice}} = 0.12$, tracing the quantitatively mild but qualitatively consistent pattern depicted in Figure 3, without requiring any additional preference parameters.

The key to explaining the divergence between the Choice and Matching methods lies in the cognitive procedure each method induces. Following Bouchouicha et al. (2025), we argue that both methods engage the same underlying cognitive mechanism, but that the procedural structure of each method determines how severely its distortions manifest. In the Matching probability method, given that the list format holds the ambiguous option fixed while varying the risky

lottery across rows, the DM is effectively induced to evaluate the fixed option first, before comparing it against a sequence of varying risky ones. This sequential structure requires the DM to form and hold an internal representation of the ambiguous event's value, which then serves as the internal anchor against which each risky option is assessed. Crucially, this first-stage evaluation is already subject to Bayesian shrinkage before any comparison takes place: the DM's noisy perception of the ambiguous probability is compressed toward the prior, amplifying a-insensitivity before any comparison with the risky option takes place. By contrast, in the Choice method, the DM evaluates each binary choice problem independently, without a pre-committed internal reference. The two options are assessed more symmetrically, allowing noise in the representations of both to partially cancel out upon direct comparison, thereby attenuating the net distortion and reducing choice noise.

Further Implications. This procedural account generates a set of sharp predictions that extend beyond the Matching–Choice comparison. Most importantly, it predicts that the Matching event method should produce a pattern that is not merely attenuated relative to MP but *reversed* in sign — constituting a-oversensitivity. Under ME, it is now the risky option that is fixed and evaluated first, while the ambiguous winning range varies across rows. By the same two-stage logic, the first-stage encoding of the fixed risky probability is subject to Bayesian shrinkage, compressing the DM's internal representation of the risky benchmark toward the prior. When the ambiguous option is subsequently compared against this compressed risky reference, a DM who exhibits ambiguity seeking for unlikely events under MP will instead appear ambiguity averse for those same events under ME, demanding a wider ambiguous winning range to compensate for the upward-compressed risky reference. This generates a prediction of systematic a-oversensitivity under the ME method: AA_p^{ME} values that are positive for unlikely events and negative for likely ones, the mirror image of the a-insensitivity pattern observed under MP.

The reversal is, however, not perfectly symmetric in magnitude. Because

the risky fixed option is encoded with higher precision than the ambiguous fixed option — a direct consequence of the greater confidence a DM has in a known probability relative to an unknown one — the first-stage anchor is less compressed under ME than under MP. The model therefore predicts that the magnitude of a-oversensitivity under ME is smaller than the magnitude of a-insensitivity under MP. The ME experiment is designed precisely to test these predictions.

4 The Matching Event: A Prospective Test

4.1 Design

The ME condition serves as a between-subject counterpart to the MP condition, with all procedural details kept identical — the same participant pool, incentive structure, comprehension checks, binary choice task, cognitive ability questions, and demographic questionnaire. The only difference lies in the format of the matching task: whereas the MP lists fix the ambiguous winning range and vary the risky winning range across rows, the ME lists fix the risky winning range and vary the ambiguous winning range across rows, as shown in Figure 4. A total of 199 participants were recruited for the ME condition.

As established in Section 2.2, the MP and ME formats partition the same 10×10 matrix of binary choice problems along orthogonal dimensions — MP groups them by columns (fixing the ambiguous option) while ME groups them by rows (fixing the risky option). The full problem set is therefore identical across the two conditions. Under any standard preference-based theory of ambiguity, this structural equivalence implies that the distribution of elicited ambiguity attitudes should be indistinguishable between the MP and ME conditions: if preferences are fixed primitives of the individual, it should make no difference whether one moves through the matrix row by row or column by column. The ME condition thus provides an unusually clean and direct test of preference-based theories — one that does not rely on between-subject differences in stimuli, incentives, or

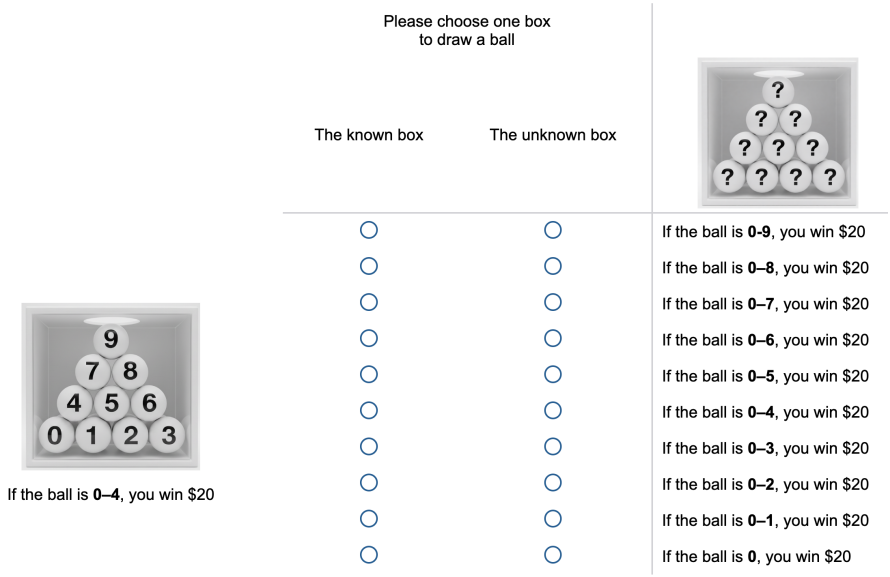

Figure 4: Screenshot of the descending matching event choice list interface. Each row presents a binary choice between the risky box (left, fixed across rows) and an ambiguous box (right) whose winning range contracts monotonically from all ten balls (0–9) at the top to a single ball (0) at the bottom. The corresponding ambiguity-neutral symmetric winning probabilities range from $p = 1$ to $p = 0.1$ in increments of 0.1.

framing. Moreover, the ME elicitation is itself a methodological contribution: to our knowledge, no prior study has implemented a matching event procedure in this structurally symmetric fashion, and the design opens a new avenue for assessing how robust measured ambiguity attitudes are to the direction of preference elicitation.

4.2 Results

Overall Pattern. To facilitate direct comparison across the two matching elicitation directions (MP *vs.* ME), we define an analogous probability-specific ambiguity attitude index for the ME elicitation. Recall that under MP, the event-specific index is defined as $AA_{\hat{p}}^{\text{MP}} = \hat{p} - mp(\hat{p})$, where $\hat{p}$ is the a-neutral likelihood of the fixed ambiguous option and $mp(\hat{p})$ is the elicited matching probability. Under ME, the roles of the two options are reversed: the risky winning probability p is fixed, and the participant's matching event $me(p)$ — the ambiguous winning range at which she is indifferent — is identified from the switch point in the list. We define

the corresponding index as

$$AA_p^{\mathrm{ME}} = \hat{p}_{me(p)} - p,$$

where $\hat{p}_{me(p)}$ denotes the a-neutral likelihood associated with the elicited matching event $me(p)$. Ambiguity aversion (seeking) again corresponds to a positive (negative) value: an ambiguity averse participant requires a wider ambiguous winning range to be willing to accept it, implying $\hat{p}_{me(p)} > p$ — the ME analog of requiring a lower matching probability under MP. Crucially, because both the event-specific $AA_{\hat{p}}^{\mathrm{MP}}$ and the probability-specific AA_p^{ME} are expressed as deviations from the a-neutral benchmark on the same probability scale, the two indexes are directly comparable across conditions, enabling a clean assessment of how the direction of preference elicitation affects measured ambiguity attitudes.

Figure 5 plots the ambiguity attitude indexes AA_p^{ME} alongside the corresponding $AA_{\hat{p}}^{\mathrm{MP}}$ indexes for reference. The ME results are strikingly consistent with the theoretical prediction: the pattern is precisely reversed relative to MP. Under ME, AA_p^{ME} is positive for small likelihoods — indicating ambiguity aversion for unlikely events — and turns negative for large likelihoods — indicating ambiguity seeking for likely ones. This constitutes the mirror image of the a-insensitivity pattern documented under MP, and we refer to it as *a-oversensitivity*: rather than compressing subjective likelihoods toward fifty-fifty, participants in the ME condition exhibit greater sensitivity to likelihood differences than the a-neutral benchmark prescribes. The reversal is not a borderline effect — it spans the entire likelihood range, and the AA_p^{ME} profile crosses zero at $\hat{p} \approx 0.5$, somewhat above the crossing point of $\hat{p} \approx 0.4$ observed under MP, roughly consistent with the symmetric two-stage account developed in Section 3.

Likelihood Sensitivity: A Quantitative Comparison. Our model further predicts that the magnitude of a-oversensitivity under ME should be smaller than the magnitude of a-insensitivity under MP, a quantitative asymmetry that rests on a foundational assumption of the noisy coding framework: that ambiguous options are inherently harder to evaluate than risky ones, and are therefore encoded

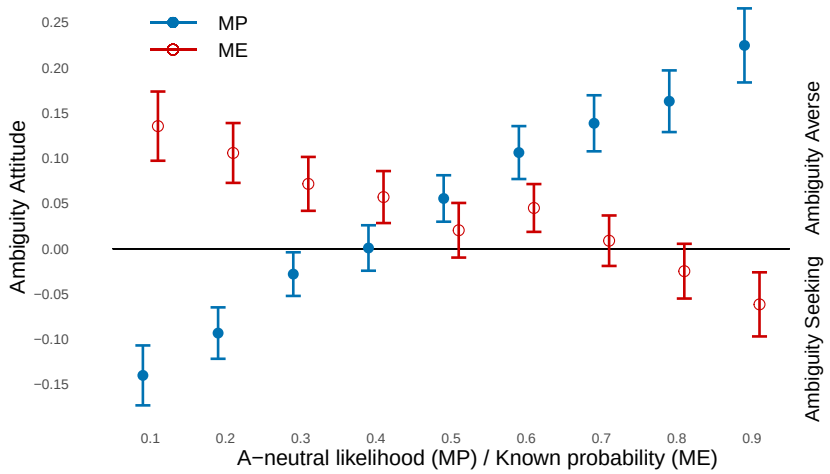

Figure 5: Event-Specific Ambiguity Attitude Indexes by Elicitation Direction. Notes: MP denotes the matching probability choice list and ME denotes the matching event choice list. Positive (negative) values indicate ambiguity aversion (seeking). Error bars denote 95% confidence intervals. See corresponding statistics in Table B.6.

with lower precision in the first stage. We find direct behavioral support for this assumption in the dominance violation data: when the fixed option in the list is degenerate, participants violate stochastic dominance significantly more often when that fixed option is ambiguous than when it is risky.[12] This indicates that participants are better able to identify the degeneracy of the fixed option when it carries a known probability than when it is ambiguous — consistent with the model's premise that ambiguous probabilities are perceived with greater noise.[13]

Inspection of Figure 5 is consistent with this prediction: the AA_p^{ME} profile is visibly flatter than its MP counterpart, with the extreme values at the tails lying closer to zero under ME than the corresponding values under MP. To quantify this asymmetry at the individual level, we first examine the individual average ambiguity non-neutrality, defined as the absolute value of the individual mean ambiguity attitude index $|\bar{AA}_i|$. A non-parametric test confirms a significantly lower level under ME than under MP (0.15 *vs.* 0.18, $p = 0.02$), providing a first

[12]As for the number of mistaken choices, 2.30 *versus* 1.28, $p < 0.01$; as for the proportion of participants who violated dominance at least once, 0.46 *vs.* 0.25, $p < 0.01$.

[13]Notably, there is no difference across the two methods in self-reported overall complexity or in time spent per list, suggesting that both types of matching lists impose similar cognitive burdens overall. The observed difference in dominance violation rates is therefore more likely to reflect the difficulty of first-stage evaluation specifically, rather than general task complexity.

piece of evidence for the prediction. We further construct a *likelihood sensitivity deviation* index,

$$\mathrm{LSD}_i = \left|AA_i(0.9) - AA_i(0.1)\right|,$$

which captures the spread of a participant's ambiguity attitude profile across the likelihood range: a larger value indicates a more pronounced deviation from a-neutrality, whether in the direction of a-insensitivity (as under MP) or a-oversensitivity (as under ME). The model predicts $\mathrm{LSD}^{\mathrm{MP}} > \mathrm{LSD}^{\mathrm{ME}}$. A non-parametric comparison confirms this: the distribution of LSD_i is significantly higher in the MP condition than in the ME condition (0.39 *vs.* 0.27, $p < 0.01$).[14]

Under any standard preference-based theory of ambiguity, the MP and ME indexes should be statistically indistinguishable: the two conditions present identical choice problems, differing only in the direction along which participants go through the same 10×10 matrix. The systematic sign reversal we observe — a-insensitivity under MP and a-oversensitivity under ME — therefore constitutes direct evidence against such theories. At the same time, the pattern aligns precisely with the predictions of the noisy coding account: the direction of the first-stage anchor determines the direction of the distortion, and the greater encoding precision of risky options relative to ambiguous ones generates the predicted asymmetry in magnitude. Taken together, the ME results do not merely replicate the MP–Choice divergence in a new format; they provide a prospective, out-of-sample validation of the two-stage mechanism, testing a prediction that no preference-based theory of ambiguity has anticipated, and one that requires specifically the two-stage structure developed here — simpler one-stage noisy coding accounts generate no such reversal.

[14]The result is robust to the alternative specification $\mathrm{LSD}_i = \left|AA_i(0.8) - AA_i(0.2)\right|$, and to a linear mixed-effects specification that covers all likelihood levels simultaneously, confirming that the magnitude asymmetry is not confined to the tails but persists across the full likelihood range.

5 Discussion

This paper examines whether measured ambiguity attitudes are robust to the method by which they are elicited. Using a within-subject design in which identical binary choice problems are presented either as standalone choices or embedded in matching probability lists, together with a between-subject design for the ME condition, we document three main findings. First, the Matching method produces substantially stronger a-insensitivity than the Choice method. Second, multiple data quality metrics — stochastic dominance violations, test-retest reliability, and between-list consistency — systematically favor the Choice method, suggesting that the stronger a-insensitivity under Matching is at least partly induced by the elicitation procedure. Third, our Matching event elicitation, in which the direction of the matching list is flipped, produces a-oversensitivity — the precise mirror image of the a-insensitivity pattern under MP — showing that the direction of the distortion is strongly affected by the structure of the elicitation method rather than by stable preferences alone. A two-stage noisy coding model, developed from Vieider (2025) and Bouchouicha et al. (2025), organizes all three sets of findings under a single cognitive mechanism and generates quantitative predictions that are broadly supported by the data.

Methodological Implications. A central practical message of this paper concerns which elicitation method researchers should use when measuring ambiguity attitudes. Our results speak to this question on multiple dimensions. On data quality grounds, the Choice method outperforms the Matching method: it produces fewer stochastic dominance violations, higher test-retest reliability, and less between-list inconsistency, all of which are indicators of a cleaner behavioral signal. At the same time, the Choice method yields attenuated a-insensitivity, raising the question of whether attenuation represents a virtue or a distortion. Our noisy coding account offers a principled answer: the Choice method recovers a less distorted ambiguity-attitude signal precisely because it does not impose a predetermined first-stage anchor, while the Matching method amplifies a-insensitivity

through the sequential list procedure. This interpretation is also consistent with recent evidence from choice-list elicitation under risk, showing that participants who fail a simple sanity test designed to assess understanding of the choice-list structure exhibit more pronounced likelihood insensitivity, even after controlling for standard comprehension checks (Qiu et al., 2025). This reinforces the view that distortions under matching are amplified by the cognitive friction imposed by the list format itself.

The sign reversal between MP and ME — both of which are matching-based methods applied to an identical problem set — further demonstrates that the measured sign of ambiguity-likelihood sensitivity is itself shaped by the elicitation direction, not exclusively a property of the decision maker. The practical implication is straightforward: researchers should favor binary choice elicitation over matching lists and other procedurally complex elicitation methods when the goal is to measure ambiguity attitudes with minimal procedural distortion. This recommendation is consistent with a broader methodological trend in the preference elicitation literature. Chapman et al. (2025) propose the Dynamically Optimized Sequential Experimentation method, which builds entirely on sequences of binary choices; they demonstrate that such procedures yield parameter estimates that are more accurate, more stable over time, and less susceptible to elicitation noise than standard matching-based approaches. Adapting such adaptive binary choice frameworks to the ambiguity domain represents a natural next step.

More broadly, the concern identified here extends beyond matching lists to any elicitation procedure that embeds a two-stage cognitive structure — including matching-based mechanisms such as BDM and willingness-to-pay or willingness-to-accept tasks, all of which require the decision maker to form an internal evaluation of a target option before reporting an equivalent value. Our results suggest that the difficulty of the first stage systematically shapes the distortion in the elicited response. In our setting, this difficulty is determined by whether the fixed option is ambiguous or risky. The second stage, however, should not be assumed to be immune: the cognitive complexity of searching through varying alterna-

tives — which may depend on the granularity of the response scale, the range of options presented, or the salience of anchor values — could further modulate the magnitude and direction of the distortion in ways that remain to be characterized. Disentangling the contributions of first- and second-stage complexity to procedural bias in preference elicitation more broadly constitutes a promising direction for future research.

The concern extends to the applied literature. Several influential studies link matching-based estimates of a-insensitivity to consequential field outcomes — stock market non-participation (Dimmock et al., 2016), portfolio underdiversification and home bias (Bianchi and Tallon, 2019), and insurance and technology adoption (Trautmann and van de Kuilen, 2015). If those estimates partly reflect procedural distortion, the magnitude of the documented associations may be sensitive to the measurement instrument, and quantitative conclusions should be interpreted accordingly. Future field studies would benefit from incorporating Choice-based elicitation, or at minimum from reporting matching-based estimates alongside a robustness check, so that the extent to which measured effects depend on elicitation procedure can be assessed.

Theoretical Implications. Our findings speak to the architecture of existing ambiguity theories in two distinct ways. First, even in our cleanest elicitation environment — the binary Choice method, the simplest pairwise comparison conceivable — we document a mild but statistically robust pattern of a-insensitivity: ambiguity seeking for unlikely events and ambiguity aversion for likely ones. This finding alone is theoretically consequential: outcome-driven models predict uniform ambiguity attitudes across the likelihood range and cannot accommodate this probability-dependent pattern (Baillon and Bleichrodt, 2015; Machina, 2014), whereas event-driven models, by operating at the level of event weighting, do so naturally. Our binary Choice results therefore weigh against the former family and lend support to the latter.

More fundamentally, the divergence across the three elicitation methods poses a challenge that goes beyond the preference-based framework entirely. Within

subjects, the same individuals exhibit mild a-insensitivity under Choice and substantially stronger a-insensitivity under MP when facing the very same 100 binary choice problems; at the between-subject level, participants assigned to the ME condition produce a-oversensitivity that mirrors the MP pattern in sign-reversed form. It is difficult to reconcile all three patterns within any framework that treats ambiguity attitudes as stable, elicitation-invariant preference parameters. Event-driven models, while capable of accommodating the binary Choice results as shown above, share with their outcome-driven counterparts the foundational assumption that ambiguity attitudes are stable primitives that elicitation merely reveals — an assumption that our cross-method evidence strongly challenges.

That said, this does not imply that these preference-based models are without value. The event-driven models in this family were, from the outset, motivated by observations about heuristic processing and systematic biases in probability judgment (Tversky and Kahneman, 1992; Tversky and Wakker, 1995; Abdellaoui et al., 2011; Baillon et al., 2025), and those insights remain highly valuable. What our results add is a sharper diagnostic: the cognitive frictions that shape ambiguity attitudes vary systematically with the procedural architecture of the elicitation task itself. The contribution of the noisy coding framework, extended here to the ambiguity domain, lies in formalizing this intuition with mathematical precision. It endogenizes ambiguity attitudes as outputs of a Bayesian encoding process whose distortions are quantitatively linked to the structure of the decision environment, and it thereby generates falsifiable predictions that our data strongly support.

References

Abdellaoui, M., Baillon, A., Placido, L., and Wakker, P. P. (2011). The Rich Domain of Uncertainty: Source Functions and Their Experimental Implementation. *American Economic Review*, 101(2):695–723.

Anantanasuwong, K., Kouwenberg, R., Mitchell, O. S., and Peijnenburg, K. (2024). Ambiguity attitudes for real-world sources: Field evidence from a large sample of investors. *Experimental Economics*, 27(3):548–581.

Baillon, A. and Bleichrodt, H. (2015). Testing Ambiguity Models through the Measurement of Probabilities for Gains and Losses. *American Economic Journal: Microeconomics*, 7(2):77–100.

Baillon, A., Bleichrodt, H., Li, C., and Wakker, P. P. (2025). Source Theory: A Tractable and Positive Ambiguity Theory. *Management Science*.

Baillon, A., Halevy, Y., and Li, C. (2022). Randomize at Your Own Risk: On the Observability of Ambiguity Aversion. *Econometrica*, 90(3):1085–1107.

Baillon, A., Huang, Z., Selim, A., and Wakker, P. P. (2018). Measuring Ambiguity Attitudes for All (Natural) Events. *Econometrica*, 86(5):1839–1858.

Belissa, T. K., Lensink, R., and Van Asseldonk, M. (2020). Risk and ambiguity aversion behavior in index-based insurance uptake decisions: Experimental evidence from ethiopia. *Journal of Economic Behavior & Organization*, 180:718–730.

Bianchi, M. and Tallon, J.-M. (2019). Ambiguity preferences and portfolio choices: Evidence from the field. *Management Science*, 65(4):1486–1501.

Bouchouicha, R., Oprea, R., Vieider, F. M., and Wu, J. (2025). Cognitive Frictions and Canonical Patterns in Risk-Taking. Working paper.

Chapman, J., Snowberg, E., Wang, S. W., and Camerer, C. (2025). Dynamically Optimized Sequential Experimentation (DOSE) for Estimating Economic Preference Parameters. Working paper.

Chateauneuf, A., Eichberger, J., and Grant, S. (2007). Choice under uncertainty with the best and worst in mind: Neo-additive capacities. *Journal of Economic Theory*, 137(1):538–567.

Dave, C., Eckel, C. C., Johnson, C. A., and Rojas, C. (2010). Eliciting risk preferences: When is simple better? *Journal of Risk and Uncertainty*, 41(3):219–243.

Dimmock, S. G., Kouwenberg, R., Mitchell, O. S., and Peijnenburg, K. (2015). Estimating ambiguity preferences and perceptions in multiple prior models: Evidence from the field. *Journal of Risk and Uncertainty*, 51(3):219–244.

Dimmock, S. G., Kouwenberg, R., and Wakker, P. P. (2016). Ambiguity Attitudes in a Large Representative Sample. *Management Science*, 62(5):1363–1380.

Einhorn, H. J. and Hogarth, R. M. (1986). Decision making under ambiguity. *Journal of business*, pages S225–S250.

Ellsberg, D. (1961). Risk, ambiguity, and the savage axioms. *The quarterly journal of economics*, 75(4):643–669.

Feldman, P. J. and Ferraro, P. J. (2025). Certain Reversals. Working paper.

Friedman, D., Habib, S., James, D., and Williams, B. (2022). Varieties of risk preference elicitation. *Games and Economic Behavior*, 133:58–76.

Frydman, C. and Jin, L. J. (2022). Efficient Coding and Risky Choice. *The Quarterly Journal of Economics*, 137(1):161–213.

Ghirardato, P., Maccheroni, F., and Marinacci, M. (2004). Differentiating ambiguity and ambiguity attitude. *Journal of Economic Theory*, 118(2):133–173.

Gilboa, I. and Schmeidler, D. (1989). Maxmin expected utility with a non-unique prior. *Journal of Mathematical Economics*, 18(2):141–153.

Holzmeister, F. and Stefan, M. (2021). The risk elicitation puzzle revisited: Across-methods (in)consistency? *Experimental Economics*, 24(2):593–616.

Kahn, B. E. and Sarin, R. K. (1988). Modeling ambiguity in decisions under uncertainty. *Journal of consumer Research*, 15(2):265–272.

Khaw, M. W., Li, Z., and Woodford, M. (2021). Cognitive Imprecision and Small-Stakes Risk Aversion. *The Review of Economic Studies*, 88(4):1979–2013.

Khaw, M. W., Li, Z., and Woodford, M. (2025). Cognitive Imprecision and Stake-Dependent Risk Attitudes. Working paper.

Klibanoff, P., Marinacci, M., and Mukerji, S. (2005). A smooth model of decision making under ambiguity. *Econometrica*, 73(6):1849–1892.

Kocher, M. G., Lahno, A. M., and Trautmann, S. T. (2018). Ambiguity aversion is not universal. *European Economic Review*, 101:268–283.

Loomes, G. and Pogrebna, G. (2014). Measuring Individual Risk Attitudes when Preferences are Imprecise. *The Economic Journal*, 124(576):569–593.

Maafi, H. (2011). Preference Reversals Under Ambiguity. *Management Science*, 57(11):2054–2066.

Machina, M. J. (2014). Ambiguity Aversion with Three or More Outcomes. *American Economic Review*, 104(12):3814–3840.

McGranaghan, C., Nielsen, K., O'Donoghue, T., Somerville, J., and Sprenger, C. D. (2024). Distinguishing Common Ratio Preferences from Common Ratio Effects Using Paired Valuation Tasks. *American Economic Review*, 114(2):307–347.

Pedroni, A., Frey, R., Bruhin, A., Dutilh, G., Hertwig, R., and Rieskamp, J. (2017). The risk elicitation puzzle. *Nature Human Behaviour*, 1(11):803–809.

Qiu, J., Wu, J., and Zhong, S. (2025). A Simple Test for Decisions under Risk. Working paper.

Savage, L. J. (1954). *The Foundations of Statistics*. Wiley, New York.

Schmeidler, D. (1989). Subjective probability and expected utility without additivity. *Econometrica*, 57(3):571–587.

Trautmann, S. T. and van de Kuilen, G. (2015). Ambiguity attitudes. In Keren, G. and Wu, G., editors, *The Wiley Blackwell Handbook of Judgment and Decision Making*, pages 89–116. Wiley-Blackwell, Oxford.

Trautmann, S. T., Vieider, F. M., and Wakker, P. P. (2011). Preference Reversals for Ambiguity Aversion. *Management Science*, 57(7):1320–1333.

Tversky, A. and Kahneman, D. (1992). Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty*, 5(4):297–323.

Tversky, A. and Wakker, P. (1995). Risk attitudes and decision weights. *Econometrica: Journal of the Econometric Society*, pages 1255–1280.

Vieider, F. M. (2024). Decisions Under Uncertainty as Bayesian Inference on Choice Options. *Management Science*, 70(12):9014–9030.

Vieider, F. M. (2025). Statistical Foundations of Ambiguity Attitudes. Working paper.

von Gaudecker, H.-M., Wogrolly, A., and Zimpelmann, C. (2025). The Distribution of Ambiguity Attitudes. Working paper.

Appendices

A A Noisy Coding Model of Ambiguity Attitude

A.1 Background: Baseline Binary Choice Under Ambiguity

We build on the statistical model of ambiguity attitudes developed by Vieider (2025). Consider a DM choosing between a risky lottery that pays a prize x with objective winning probability p_r and an ambiguous lottery that pays the same prize with unknown winning probability. The DM encodes her beliefs about each winning probability as a Beta distribution: $p_a(x) = \mathrm{Be}(\alpha_a, \beta_a)$ for the ambiguous option and $p_r(x) = \mathrm{Be}(\alpha_r, \beta_r)$ for the risky option. Both distributions share the same mean belief p, reflecting the symmetry of the Ellsberg urn design, but differ in concentration: $\kappa_a \triangleq \alpha_a + \beta_a < \kappa_r \triangleq \alpha_r + \beta_r$. This difference in concentration captures the notion that the DM is less confident in her probability assessment for the ambiguous urn than for the risky urn, formalizing ambiguity as a lack of confidence rather than as a distinct preference parameter.

Log-odds representation. It is computationally convenient to work on the log-odds scale (Vieider, 2025). Define

$$\ell(p) \triangleq \log \frac{p}{1-p}, \qquad \mu_0 \triangleq \ell(p_0),$$

where p_0 is the DM's prior reference probability. The precision of the log-odds signal for each urn follows from the Beta concentrations via the Trigamma function: $\lambda_a \triangleq [\psi'(\alpha_a) + \psi'(\beta_a)]^{-1}$ and $\lambda_r \triangleq [\psi'(\alpha_r) + \psi'(\beta_r)]^{-1}$, with $\lambda_a < \lambda_r$ reflecting the lower precision of the ambiguous signal. Let ξ denote the precision of the prior. The Bayesian posterior weights are then

$$\rho \triangleq \frac{\lambda_r}{\lambda_r + \xi}, \qquad \gamma \triangleq \frac{\lambda_a}{\lambda_a + \xi}$$

Both ρ and γ capture the weight placed on the perceived signal relative to the prior mean; $0 < \gamma < \rho < 1$ because the ambiguous signal is less precise and therefore receives less weight.

Binary choice. In a standalone binary choice between the two options, the DM evaluates each option independently and simultaneously. The log-odds of the risky and ambiguous options are each shrunk toward the prior, and the DM selects the option with the higher posterior. Stochastic indifference in binary choice is characterized by the equivalence between the posterior mean for these two options:

$$\rho\,\ell(\overline{mp}) + (1-\rho)\mu_0 = \gamma\,\ell(p_a) + (1-\gamma)\mu_0,$$

where $\overline{mp}$ denotes the matching probability under the binary choice format for the ambiguous event with a-neutral probability p_a. Thus, the binary matching probability satisfies

$$\ell(\overline{mp}) = \mu_0 + \frac{\gamma}{\rho}\big[\ell(p_a) - \mu_0\big].$$

Since $0 < \gamma/\rho < 1$, the binary matching probability is a compressed transformation of the a-neutral probability on the log-odds scale: low ambiguous probabilities are pulled upward toward the prior and high ambiguous probabilities are pulled downward, generating a-insensitivity endogenously.

Prediction 1 (Binary a-insensitivity)**.** *In binary choice, the matching probability is compressed toward the prior:*

$$\ell(\overline{mp}) - \mu_0 = \frac{\gamma}{\rho}\big[\ell(p_a) - \mu_0\big], \qquad 0 < \frac{\gamma}{\rho} < 1.$$

Corollary to Prediction 1 (Binary choice symmetry)**.** *In a standalone binary choice, the DM evaluates both the risky and the ambiguous option simultaneously and independently, with no pre-committed anchor. The indifference condition is therefore symmetric: whether the task is framed as finding a matching probability for a fixed ambiguous event or as finding a matching event for a fixed risky probability, the underlying comparison is identical. Formally, the ME indifference*

condition in binary choice satisfies

$$\ell(p_r) - \mu_0 = \frac{\gamma}{\rho}\big[\ell(\overline{me}) - \mu_0\big].$$

which is the exact mirror of the MP condition in Prediction 1, where $\overline{me}$ *denotes the a-neutral probability of matching event under the binary choice format for the risky event with probability* p_r*. The two framings therefore predict identical ambiguity attitude indexes in binary choice:* $\overline{AA}_{MP} = \overline{AA}_{ME}$ *for all likelihood levels, with ambiguity aversion for unlikely events and ambiguity seeking for likely ones. This symmetry is broken once the binary choices are embedded into a list, as the two-stage procedure introduces an asymmetric first-stage anchor whose direction—ambiguous under MP, risky under ME—determines whether the resulting attitude index is compressed or expanded relative to the binary benchmark.*

A.2 Extension: Two-Stage List Logic

We extend the baseline model to the list environment following Bouchouicha et al. (2025). Both the MP and ME list formats induce a two-stage cognitive procedure: the DM first evaluates the *fixed* option to form an internal anchor, and then searches the varying rows for the option whose signal best matches that anchor. Because both sides of the comparison are probabilities, the model stays entirely on the log-odds scale throughout, making the present extension structurally cleaner than the original certainty-equivalent application in Bouchouicha et al. (2025).

Stage 1: evaluating the fixed option. Let θ denote the latent log-odds of the fixed option, and let the internal signal be

$$s \mid \theta \sim N(\theta,\, \lambda^{-1}), \qquad \theta \sim N(\mu_0,\, \xi^{-1}).$$

The stage-1 posterior mean is

$$F(s) = \frac{\lambda}{\lambda+\xi}\, s + \frac{\xi}{\lambda+\xi}\, \mu_0,$$

with conditional posterior variance $V = (\lambda+\xi)^{-1}$. For a risky fixed option, the posterior weight is ρ and the posterior variance is $V_r = (\lambda_r+\xi)^{-1}$; for an ambiguous

fixed option, the posterior weight is γ and the posterior variance is $V_a = (\lambda_a + \xi)^{-1}$.

Stage 2: searching the varying rows. The DM searches for the varying row whose signal best matches the stage-1 anchor F.

A.2.1 Matching Probability List (MP)

In a MP list, the fixed option is ambiguous with a-neutral probability p_a, and the varying options are risky probabilities $q_1 < q_2 < \cdots < q_J$.

Stage 1. The stage-1 posterior mean centers at

$$\mathbb{E}[F_a] = \mu_0 + \gamma\big[\ell(p_a) - \mu_0\big].$$

At this stage, the ambiguous signal is compressed toward the prior to the same extent as under the binary choice format.

Stage 2. The DM selects the risky row whose signal best matches $\mathbb{E}[F_a]$. In a continuous approximation, the latent MP response centers at

$$\ell(\widetilde{mp}) = \mu_0 + \gamma\big[\ell(p_a) - \mu_0\big].$$

At this stage, the effective precision of the matched signal combines coding noise on the varying side with uncertainty inherited from the stage-1 posterior:

$$\tilde{\lambda}_r \triangleq \left(\lambda_r^{-1} + V_a\right)^{-1} = \left(\lambda_r^{-1} + (\lambda_a + \xi)^{-1}\right)^{-1}.$$

Prediction 2 (MP a-insensitivity)**.** *The central MP response implies*

$$\ell(\widetilde{mp}) - \mu_0 = \gamma\big[\ell(p_a) - \mu_0\big], \qquad 0 < \gamma < 1.$$

Prediction 3 (Stronger compression under MP than binary choice)**.** *Since* $0 < \rho < 1$*, we have* $0 < \gamma < \gamma/\rho < 1$*, so MP produces stronger a-insensitivity than standalone binary choice.*

A.2.2 Matching Event List (ME)

In a ME list, the fixed option is risky with probability p_r, and the varying options are ambiguous events with best-guess probabilities $e_1 < e_2 < \cdots < e_J$.

Stage 1. The stage-1 posterior mean centers at

$$\mathbb{E}[F_r] = \mu_0 + \rho\big[\ell(p_r) - \mu_0\big].$$

The risky signal is less compressed than the ambiguous signal since $0 < \gamma < \rho < 1$.

Stage 2. The DM selects the ambiguous row whose signal best matches $\mathbb{E}[F_r]$. In a continuous approximation, the latent ME response centers at

$$\ell(\widetilde{me}) = \mu_0 + \rho\big[\ell(p_r) - \mu_0\big].$$

The effective stage-2 precision is

$$\tilde{\lambda}_a \triangleq \left(\lambda_a^{-1} + V_r\right)^{-1} = \left(\lambda_a^{-1} + (\lambda_r + \xi)^{-1}\right)^{-1}.$$

Prediction 4 (ME a-oversensitivity)**.** *The central ME response satisfies*

$$\ell(p_r) - \mu_0 = \frac{1}{\rho}\big[\ell(\widetilde{me}) - \mu_0\big], \qquad 0 < \rho < 1.$$

Since $1/\rho > 1$, the ME-list induces oversensitive ambiguity attitudes—a reversed pattern relative to both the binary choice and the MP-list—ambiguity aversion for unlikely events and ambiguity seeking for likely events.

Prediction 5 (Stronger compression under MP than ME)**.** *The insensitivity induced by the MP-list can be proxied by $1 - \gamma$, whereas the oversensitivity induced by the ME-list can be proxied by $1 - \rho$. Since $0 < \gamma < \rho < 1$, the model predicts stronger prior-centered compression under MP than under ME. This asymmetry arises because MP anchors on the ambiguous fixed option in stage 1, whose signal is more heavily shrunk toward the prior, whereas ME anchors on the risky fixed option, whose signal is less shrunk.*

B Additional Figures and Tables

Table B.1: Event-Specific Ambiguity Attitude Indexes by Elicitation Method

	A-neutral likelihood $\hat{p}$								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
Matching	−0.140	−0.093	−0.028	0.001	0.056	0.106	0.139	0.163	0.225
Choice	−0.062	−0.044	−0.012	0.011	0.038	0.060	0.088	0.108	0.121
p-value	<0.001	<0.001	0.197	0.399	0.189	0.004	0.002	0.003	<0.001

Notes: Each cell reports the mean $AA_{\hat{p}} = \hat{p} - mp(\hat{p})$ for the indicated elicitation method and a-neutral likelihood $\hat{p}$. Positive (negative) values indicate ambiguity aversion (seeking). p-values are from two-sided paired t-tests of the null hypothesis that $AA_{\hat{p}}$ is equal across Matching and Choice.

Table B.2: Response frequency by event likelihood and response category.

	A-neutral probability $\hat{p}$								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
$AA^{\text{Matching}} < AA^{\text{Choice}}$	50	62	59	50	63	78	84	85	93
$AA^{\text{Matching}} = AA^{\text{Choice}}$	68	57	63	74	75	65	59	44	41
$AA^{\text{Matching}} > AA^{\text{Choice}}$	83	82	79	77	63	58	58	72	67

Notes: $\hat{p}$ denotes the event's A-neutral probability. AA^{Matching} represents the AA derived from Matching responses, while AA^{Choice} represents the AA derived from Choice responses. $N = 201$ per $\hat{p}$ level.

Table B.3: Classification of participants by preference reversal type across elicitation methods and A-neutral probability level.

	A-neutral probability $\hat{p}$									
Group	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
No reversal	129	130	135	135	132	138	143	142	151	134
PR1	41	48	42	48	42	38	35	42	34	48
PR2	31	23	24	18	27	25	23	17	16	19
PR (%)	35.8	35.3	32.8	32.8	34.3	31.3	28.9	29.4	24.9	33.3

Notes: No reversal denotes participants whose AA carries the same sign across both methods (both positive or both negative). PR1 represents the type-I preference reversal and denotes participants who are ambiguity seeking under Matching but ambiguity averse under Choice ($AA^{\text{Matching}} < 0 < AA^{\text{Choice}}$). PR2 represents the type-II preference reversal and denotes participants who are ambiguity averse under Matching but ambiguity seeking under Choice ($AA^{\text{Choice}} < 0 < AA^{\text{Matching}}$).

Table B.4: Regression results for a-insensitivity and ambiguity aversion indices.

	A-insensitivity (a)		Ambiguity aversion (b)	
	(1)	(2)	(3)	(4)
Intercept	0.23^{***}	0.29^{***}	0.07^{***}	-0.05
	(0.03)	(0.10)	(0.02)	(0.07)
Matching	0.20^{***}	0.20^{***}	0.03	0.03
	(0.03)	(0.03)	(0.02)	(0.02)
Order (DESC)		0.02		0.01
		(0.04)		(0.03)
Age		0.00		0.00
		(0.00)		(0.00)
Female		0.06		0.05^{*}
		(0.04)		(0.03)
Cognition		-0.01^{*}		0.00
		(0.01)		(0.00)
$\Delta_{Complexity}$		-0.00		-0.02
		(0.02)		(0.02)

Notes: All models are estimated with random intercepts for participants. The variable Matching is a dummy equal to 1 for observations elicited via the Matching method. $\Delta_{Complexity}$ denotes the difference in perceived complexity between the two elicitation formats. Standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table B.5: Test-retest reliability of elicitation methods by known probability level.

	Switching Frequency			Cohen's κ	
$\hat{p}$	Matching	Choice	Fisher p	Matching	Choice
0.1	33	24	0.253	0.580	0.452
0.2	48	43	0.634	0.515	0.573
0.3	38	35	0.796	0.503	0.439
0.4	37	13	0.000	0.406	0.571
0.5	29	14	0.023	0.372	0.463
0.6	18	7	0.037	0.558	0.517
0.7	16	6	0.046	0.556	0.557
0.8	19	6	0.012	0.461	0.556
0.9	19	3	0.001	0.405	0.762

Notes: $N = 201$ per $\hat{p}$ level. Switching rate denotes the number of participants whose choice reverses between the two repetitions. Fisher p denotes the p-value from Fisher's exact test comparing switching rates across methods. Cohen's κ measures agreement between the two repetitions within each method, corrected for chance.

Table B.6: Mean Ambiguity Attitude Index across Likelihood Levels by Elicitation Task

	A-neutral likelihood (MP) / Known probability (ME)								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
MP	−0.140	−0.093	−0.028	0.001	0.055	0.106	0.139	0.163	0.225
ME	0.135	0.106	0.072	0.057	0.020	0.045	0.009	−0.025	−0.062

Notes: For the MP task, each cell reports the mean $AA_{\hat{p}}^{MP} = \hat{p} - mp(\hat{p})$, where e denotes the A-neutral likelihood of the ambiguous lottery and $mp(e)$ is the elicited matching probability of the risky lottery. For the ME task, each cell reports the mean $AA_{p}^{\mathrm{ME}} = \hat{p}_{me(p)} - p$,, where p denotes the known probability of the risky lottery and $mp(p)$ is the elicited matching probability of the ambiguous lottery. Positive (negative) values indicate ambiguity aversion (seeking).